CAMBRIDGE
International Education

Cambridge IGCSE™

FOOD AND NUTRITION **0648/12**

Paper 1 Theory **May/June 2026**

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **27** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states ‘show your working’. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	**Meaning**
✓	Correct point or mark awarded
✗	Incorrect point or mark not awarded
BOD	Benefit of the doubt given
Highlighted text	Highlighting areas of text
On-page comment box	Allows comments to be entered on the page
Off-page comment box	Allows comments to be entered at the bottom of the marking window and then displayed when the associated question item is navigated to
^	Information missing or insufficient for credit
REP	Repetition in response
I	Incorrect or insufficient point ignored while marking the rest of the response

Annotation	Meaning
R	Incorrect point or mark not awarded
CON	Contradiction in response, mark not awarded
SEEN	Point has been noted, but no credit has been given or blank page seen
	Key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen

Question	Answer	Marks	Guidance
1(a)	*what is meant by DRV* Dietary Reference Value;	1	All three words must be correct to obtain mark **A** alternative phonetic spelling
1(b)	*reasons a list of ingredients on a food label is useful to a consumer* can compare one product with another / make informed choice / can see exactly what product is made from/contains e.g. additives / can tell what product will taste like / can see amount of ingredients that have been used; can see if any ingredients are/are not compatible with preference likes / dislikes e.g. food types, additives; can see if any ingredients are / are not compatible with allergies; can see if any ingredients are / are not compatible with intolerances; can see if ingredients comply with dietary needs for health reasons e.g. diabetes, heart issues, weight issues; can see if ingredients comply with dietary needs for ethical / moral purposes e.g. vegetarian; can see if ingredients comply with specific dietary need for religious purposes; can see main ingredient and know what they are paying for (as ingredients listed in descending order of quantity);	3	Any three

Question	Answer	Marks	Guidance
2(a)(i)	*mineral that is found in milk* calcium / phosphorus / potassium / iodide/iodine;	1	
2(a)(ii)	*deficiency disease associated with a lack of mineral named in **(a)(i)*** calcium – rickets / tetany / osteomalacia / osteoporosis; phosphorus – rickets / tetany / osteomalacia / osteoporosis; potassium – hypokalaemia; iodide / iodine – goitre;	1	**ECF** if incorrect mineral named in **(a)(i)** but a correct deficiency named for given mineral, then award mark here

Question	Answer	Marks	Guidance
2(b)(i)	*mineral that helps control the rate of metabolism* iodide / iodine; *gland that controls the level of energy* thyroid gland;	**2**	1 × 1 mark for mineral 1 × 1 mark for gland
2(b)(ii)	*factors that affect energy requirements* activity / PAL / lifestyle / exercise; activity of the thyroid gland; age; biological sex; body mass / size / weight / height / BMI; climate; lactation / breastfeeding; occupation / profession; pregnancy; state of health / illness / after surgery / convalescence / after birth; stress level;	**4**	Any four factors **A** gender

Question	Answer	Marks	Guidance
2(b)(iii)	*effects on the body when energy intake is less than energy output* body becomes / feels cold; death; dizziness / faintness / headache; during pregnancy increased risk to unborn baby / risk of low birth weight baby; hair loss / thinning hair; lack of growth (children) / stunting / wasting; muscle wastage / emaciated; problems producing milk during lactation; recovery slowed down during illness; reduced energy levels / lethargy / tiredness / fatigue / weakness; reduced or stopped menstrual cycle / infertility; risk of becoming anorexic; weight loss / underweight / energy used from fat stores so these are depleted;	**4**	Any four effects
2(b)(iv)	*example of a process carried out by the body that uses mechanical energy* muscle movement, voluntary and involuntary / work (or examples) / any physical activity e.g. running, jumping, walking, chewing, blood circulation, peristalsis, breathing, blinking, respiration, digestion, metabolism;	**1**	

Question	Answer	Marks	Guidance
3(a)	*different foods that are a good source of starch* breakfast cereals or one named example; cereals or one named example e.g. oats, barley, rye, rice, maize / corn; grains – amaranth, buckwheat, millet, quinoa (aka kaniwa), sorghum, teff; legumes / pulses or one named example e.g. beans, peas, lentils; plantain / banana; root vegetables or one named example e.g. parsnips, rutabaga, sweet potatoes, yuca (aka cassava aka tapioca), taro root; sago; tubers or one named example e.g. potato, yam, Jerusalem artichoke;	3	Any three **different** named food sources **R** anything made with wheat / wheat flour
3(b)(i)	*enzyme that begins the breakdown of starch* (salivary) amylase / ptyalin;	1	**R** pancreatic amylase, carbohydrase
3(b)(ii)	*where in the digestive system breakdown of starch begins* in mouth / salivary glands;	1	

Question	Answer	Marks	Guidance
4(a)	*symptoms of pellagra* diarrhoea; dermatitis; dementia;	2	Any two symptoms **A** alternative phonetic spellings
4(b)	*other function of vitamin B_3* involved in release of energy from food / protein / fat / carbohydrate; healthy skin; nerve function;	1	**R** pellagra prevention

Question	Answer	Marks	Guidance
4(c)	*sources of vitamin B_3 for vegetarians* avocado; fortified cereal products e.g. breakfast cereals, bread; grains e.g. amaranth, buckwheat, millet, quinoa, sorghum, teff; milk and dairy foods; mushrooms; nuts; peanuts; potatoes; pulses e.g. beans, peas, lentils; seeds e.g. sunflower, chia, pumpkin; wheat germ; wholegrain / wholemeal cereals e.g. sweetcorn, brown rice, oats; yeast and yeast extracts / nutritional yeast / Marmite;	**4**	Any four different suitable foods

Question	Answer	Marks	Guidance
5(a)	*differences between fats and oils* fats are generally saturated (more single bonds between its carbon atoms) compared to oils which are mostly unsaturated; fats usually have a higher melting point than oil / oil has a lower melting point than fat; fat can originate from a plant and / or an animal source – oil originates from a plant and / or a fish source;	**2**	differences must be clearly given to gain full marks

Question	Answer	Marks	Guidance
5(b)	*functions of vitamin D* growth; helps calcium absorption / helps achieve peak bone mass / keeps bones healthy; <u>maintenance</u> of bones / teeth; prevents (onset of) osteoporosis (which increases risk of bone fractures); prevents osteomalacia (in adults); prevents rickets (in children); promotes quicker healing of bone fractures; required for blood clotting; (works with calcium and phosphorous to) <u>forms / builds</u> strong teeth and bones;	**4**	Any four functions **R** strengthen bones/teeth

Question	Answer	Marks	Guidance
6	*reasons why protein foods are an important part of a balanced diet* many substances attach themselves to protein to facilitate transport in the body, e.g. haemoglobin is a transporting protein, to carry oxygen, help with absorption after digestion; prevention of extreme conditions such as kwashiorkor / marasmus; producing antibodies (protective proteins produced by the immune system) which attach to antigens (foreign substances) — such as bacteria, fungi, viruses and toxins — and remove them from the body; production / formation of enzymes which facilitate most of the chemical reactions in the body / vital for metabolism including digestion of nutrients, regulation of energy production in cells; production of hormones which can regulate metabolic rate, blood glucose levels; proteins form part of the structural framework of bones; structure of the body – each cell contains protein as part of the cell membrane; to maintain / replace / renew / repair / body / cells as they wear out; to supply a <u>secondary</u> source of energy if not enough carbohydrate or fat available;	**5**	Any five reasons

Question	Answer	Marks	Guidance
7(a)	*reasons for using plain white flour to make shortcrust pastry* plain flour gives a short crumb texture / crumbly result (as pastry is less elastic / stretchy); plain flour has no raising agent (so it prevents pastry rising / gives traditional flat appearance); plain flour is a soft flour / has a <u>low</u> gluten content; white flour as wholemeal flour is heavy and gives a denser result;	**2**	Any two suitable reasons **R** soft / light texture **R** any reference to flavour

Question	Answer	Marks	Guidance
7(b)	*reason why butter is used to make shortcrust pastry* butter is a hard fat which does not melt when rubbing in; gives pastry (a good) flavour; gives pastry (a good) colour;	**2**	Any two suitable reasons
7(c)	*proportion of fat to flour used when making shortcrust pastry* ½ / half / 1:2;	**1**	**A** flour to fat 2:1 if clearly specified
7(d)	*method used to make shortcrust pastry* rubbing-in;	**1**	
7(e)	*method of incorporating air when making shortcrust pastry* sieving the flour / rubbing in the fats;	**1**	
7(f)	*reasons why a pastry case is baked blind* allows case to hold its shape as gluten / protein in flour coagulates; ensures pastry crust/base is fully cooked / baked properly / cooked thoroughly; give a stronger pastry case; gives a crisp / short / crumbly texture to the pastry (due to dextrinisation); prevent pastry rising through the filling; prevents leaking of filling through pastry; the filling has a shorter cooking time than the pastry / pastry needs longer cooking time than filling / ensures filling not over-cooked; to avoid the unbaked and / or wet filling from making the pastry soggy / oily / moist / soft; used when the filling is uncooked;	**4**	Any four reasons

Question	Answer	Marks	Guidance
7(g)	*reasons why the pastry may shrink during baking* incorrect / low oven temperature / constantly opening oven door; not allowed pastry to rest after rolling / shaping before baking (to relax the gluten); not rolled with short, sharp strokes in a forward direction / pastry over-stretched during rolling out / rolled too much; not supported pastry on rolling pin when lifting; over rubbing-in during making (which strengthens gluten); over-kneading during making (which strengthens gluten); pastry not turned round for even rolling; stretched during shaping / lining flan ring etc.;	**4**	Any four reasons
7(h)	*reasons why a consumer may use pre-prepared pastry* can be stored for emergencies as frozen / packet mix; cheaper / cost effective / may be cheaper than buying all ingredients to make from scratch; consistent quality / reliable results / guaranteed result / eliminates errors; consumer prefers taste/texture of pre-pared pastry to home-made; different varieties available so more variety cooking for family; easier / good for those who do not like cooking / useful for people who do not have skill to make pastry / pastry is too complicated to make; no waste of ingredients being left over; nutritional information on packaging;	**4**	Any four reasons **R** any reference to saving time in any way e.g. shopping for ingredients / preparing / washing up

Question	Answer	Marks	Guidance
8	*reasons why some foods are preserved by irradiation* destroys enzymes that cause ripening of fruit / vegetables; keeps flavour / texture of food / little/minimal change / loss in flavour / texture; makes food safer to eat / prevents food poisoning by destroying microorganisms; prevents insect infestation in dry foods and food products; reduces the need for chemical preservatives; reduces the numbers of pathogenic parasites in meat; reduces / slows down food spoilage / decay by mould, yeast, bacteria; slows down sprouting of vegetables like potatoes and onions; transportation of products easier and safer;	**4**	Any four reasons **R** any reference to nutritional content

Question	Answer	Marks	Guidance
9(a)	*advantages of silicone as a material for bakeware* available in different colours / attractive; can be bought in many shapes e.g. muffin tray, sponge tin; does not rust; durable / long-lasting / re-usable; easy to clean / dishwasher safe; flexible so does not get dented or chipped if dropped; freezer safe; heat resistant / poor conductor of heat / good insulator / conducts heat evenly; lightweight; microwave safe; non-stick for easy removal of food; non-toxic / non-reactive to pH of food / odour free so does not affect quality of baked food; not very expensive / affordable / relatively cheap;	**5**	Any five advantages **A** does not contain microplastics

Question	Answer	Marks	Guidance
9(b)	*materials that can be used for bakeware* (cast) iron; aluminium / foil; ceramic / clay / stone; enamel / vitreous enamel; glass; ovenable paperboard; stainless steel / (carbon) steel / (aluminised) steel / (tin-plated) steel;	**2**	Any two materials **R** silicone / rubber, copper, tin **A** metal in place of named metal once only

Question	Answer	Marks	Guidance
10(a)	*ways to safely dispose of kitchen waste to prevent infestation from vermin* cover kitchen/outside bin tightly / use secure bins / use bins with a lid; empty all kitchen bins regularly / do not allow kitchen / outside bin to overflow / put full kitchen bin in outside bin immediately / do not leave full waste bins in kitchen; keep outside bin area clean; keep outside bin away from house; keep separate bin for food waste / compost food waste away from the kitchen area; line bins with plastic bin liner / wrap waste food; put all kitchen waste in kitchen bin immediately / do not allow kitchen waste to accumulate on surfaces; tie bin liners / bags; use a kitchen waste disposal unit; wash / disinfect all bins regularly;	**4**	Any four relevant ways

Question	Answer	Marks	Guidance
10(b)	*symptoms of food poisoning* abdominal pain / stomach cramps; backache; coma; confusion; dehydration; diarrhoea / upset stomach; dizziness; double vision; exhaustion / tiredness / fatigue; fever / shivery / high temperature / flu-like symptoms; headache; kidney failure; loss of appetite; miscarriage / still births / illness in new born baby; pneumonia; rash; septicaemia; sweating; vomiting / sickness / nausea;	**3**	Any three

Question	Answer	Marks	Guidance
11(a)	*how the cook-chill process helps to prevent food spoilage* foods selected for cook-chill are of high quality; foods are prepared / cooked in a hygienic / sterile professional kitchen / factory; foods are cooked to a high temperature to destroy food spoilage organisms / yeast / mould / bacteria; foods are cooked to a high temperature to destroy enzymic activity that causing ripening so leading to food spoilage; cooked foods are packaged in sterile / hygienic containers to prevent any contamination; cooked food is chilled / cooled quickly / immediately / rapidly through the danger zone temperature (from +70 °C to 8 °C within 90 minutes); foods are chilled to remove heat so preventing / slowing down growth of microorganisms; foods are chilled to remove heat to slow down enzymic activity; foods are stored out of the danger zone / at a temperature between 0 °C and 8 °C / in the chiller / refrigerator; foods are packaged / sealed to prevent entry of microorganisms / dust / vermin;	**4**	Any four

Question	Answer	Marks	Guidance
11(b)	*disadvantages of cook-chill products* food must be refrigerated and this may be a problem for those people with limited access to refrigeration; foods cannot be eaten immediately / not ready to eat / food requires reheating; if food is not reheated thoroughly/ until piping hot (until centre reaches at least 72 °C for two minutes) consumer may suffer from food poisoning; foods can only be kept for a few days / have a shorter shelf-life than frozen foods; short shelf-life may make them unsuitable for bulk purchase so frequent shopping trips are required incurring costs in time/fuel; products require a lot of packaging which may be unacceptable to consumers concerned about environmental issues; if using cook-chill for a number of people, then it is more expensive than cooking from raw ingredients; foods can only be reheated once;	4	Any four disadvantages **R** any mention of texture, flavour, nutrient destruction, additives, portion size Reference must be clear that it is expensive to purchase cook-chill products when catering for several people

Question	Answer	Marks	Guidance
12	*Discuss health and safety guidelines that should be followed in the kitchen to prevent falls, burns and scalds.* *falls [max 8]* avoid overfilling pots with water / oil / food to minimise the chances of it boiling over as spilt liquid / food may result in a fall; do not run / rush around in the kitchen as this could result in a fall; do not wear long / loose / trailing clothing while cooking as these can be a trip hazard; floor should be evenly laid with no worn / broken flooring with holes or cracks to prevent catching foot in and falling; floor should be non-slip / textured / not highly polished as this may result in falls; floor should not have loose mats to prevent catching foot in and falling; if having to use items above easy reach / high places use kitchen steps to avoid falls; keep equipment where it can easily be reached / do not keep heavy items in high cupboards to avoid climbing on surfaces or over-balancing resulting in a fall; keep hair tied back / covered to ensure good visibility whilst working; keep kitchen well-ventilated to allow heat / smoke / steam to escape and prevent obscuring of vision / fainting in heat or a fall could happen;	**15**	Credit AVP. Must demonstrate sound understanding of topic for full marks e.g. covers each area of the question equally can provide relevant explanation and / or examples for each statement where appropriate shows good knowledge and understanding of guidelines to follow to prevent falls shows good knowledge and understanding of ways to prevent burns and scalds correct terminology used where appropriate comments precise and relevant information should be presented in a clear and organised way

Question	Answer	Marks	Guidance
12	keep pets / children out of the kitchen to ensure they do not become a trip hazard; make sure lighting is good to ensure clear vision and avoid obstacles that could cause fall; keep cupboard / oven doors closed / make sure that all trip hazards / obstructions / furniture is not cluttering walkways to prevent a fall; no trailing cables / flexes from equipment / use coiled flexes to stop wires dangling to prevent tripping and falling over; take care when using water / washing up to avoid spilling water / wipe up spilled liquid immediately and dry area as this could cause a fall; wear suitable footwear tie shoelaces / avoid high heels / open sandals / slippers as these can cause falls; wipe up grease / food debris on floors immediately to avoid slipping and falling; *burns and scalds [max 8]* allow all hot pans / bakeware / stove tops to cool prior to washing to prevent burns; cover frying pans with frying screens / splatter guard or lids to prevent spitting causing scald; do not have bare feet / wear open sandals as these provide no protection from scalds or burns if hot liquid or hot items dropped; do not have flames too high when heating oil as this could ignite and cause burns; do not reach over a pan on front burner to get to back burner to prevent scald and danger of clothing catching fire from gas flame;		

Question	Answer	Marks	Guidance
12	do not use electrical equipment or turn on plugs with wet hands or use near water or overload sockets as this may result in electrical burn; do not use metal tools / plates in microwave / toaster as this could result in electrical burn; do not use water to extinguish a fire caused by an electrical fault as water is a conductor of electricity and this could result in an electrical burn; have a flat base on frying pan / saucepan so it sits securely on burner so less risk of it falling and hot liquid causing scald; do not overfill kettle / keep kettle spouts away from edge of work-surface to avoid scalding with steam; tie back long hair when cooking as it could catch fire; turn all pan handles away from edge of stove / ensure handles are not overheat source to avoid burning / use stove guard to prevent knocking pans down and scalding; ensure all pan handles are made from insulated material to prevent burns; do not leave boiling / frying pans unattended when cooking due to danger of contents boiling over / igniting and causing burns / scalds; turn off hot plates/hobs when not in use to prevent accidental burns; use oven gloves not tea-towels for hot pans/dishes or hands may get burnt or scalded or dish may drop and burn feet; make sure oven gloves are dry or scalds can occur due to heat penetration; use tongs to turn food not fingers to prevent burns / do not use metal spoons for stirring or leave them in pan as they conduct heat and cause burns; wear suitable clothing whilst cooking, avoid loose fitting sleeves dangling scarves to prevent them catching fire and causing burns; when boiling / frying do not overheat water / oil as the liquids could boil over and cause a scald;		

Question	Answer	Marks	Guidance
12	when boiling / frying / using kettle do not overfill with water / oil / food due to danger of overflowing and scalding; when frying if pan catches fire do not move pan until fat is cold or it may catch fire again and scald hand whilst moving; when frying lower food gently into fat to avoid splashing fat and causing burns; when frying make sure pan / equipment / food is dry as water turns to steam and splutters causing burns; when using a food processor with hot liquid do not over fill as it may cause scalding; when opening microwave / lidded pan / pressure cooker / steamer / opening oven door keep face safe distance away to avoid scalding with steam; when draining boiling water from pan at the sink take care not to scald face / hands; if using a blow torch take extreme care to prevent burns; if removing hot pans / bakeware from hob / oven warn others to avoid accidents which may cause burns / scalds;		

Question	Answer	Marks	Guidance
13	*Discuss how the nutritional benefits and versatility of eggs make them a popular commodity.* *nutritional benefits [max 8]* protein – contains all EAA, is HBV, growth, repair, maintenance, secondary energy, hormones, enzymes, antibodies; fat – insulation, contains EFA's, warmth, energy, protection internal organs, supplies fat-soluble vitamins A, D, E, K; vitamin A / retinol – antioxidant, destroys free radicals / helps prevent risk of cancer, promotes healthy skin, forms mucous membranes in throat / digestive / bronchial / excretory tracts, healthy immune system / helps prevent infection, helps vision in dim light / at night. prevents night blindness / xerophthalmia, production of visual purple / rhodopsin in retina of eye, required for growth / embryonic development, helps keep mucous membranes moist and free from infection; vitamin D / cholecalciferol – growth, helps absorption of calcium / phosphorous, maintenance of bones / teeth, prevents onset of osteoporosis which increases risk of bone fractures, prevents osteomalacia in adults a disease resulting in calcium loss and softening of bones, prevents rickets in children where bones are weak and malformed with bow legs, knock-knees, enlargement of bones, promotes quicker healing of bone fractures, (works with calcium and phosphorus to) forms / builds bones / teeth; vitamin B2 / riboflavin – release energy from carbohydrates, amino acids and fats, growth, nerve function, healthy clear skin, prevents beri-beri; vitamin B9 / folate – normal growth, formation of red blood cells, releases energy from amino acids, production of DNA and RNA, protects against neural tube defects such as spina bifida in unborn babies;	**15**	Credit AVP. must demonstrate sound understanding of topic for full marks e.g. covers each area of the question equally can provide relevant explanation and / or examples for each statement where appropriate shows good knowledge and understanding of nutritional benefits of eggs Should name nutrient and 2 functions for full mark If only named nutrients with no function then ● and allow 1 mark for 2 nutrients [vitamin B1 / thiamine and vitamin B3 /niacin amounts very limited, if said B-group vitamin / B1 / B3 and given 2 functions then can have mark but not if also given B2, B9, B12)] **A** zinc – antioxidant, helps immune system, repairs wounds, enzyme function, bone development, production DNA, aids fertility, blood clotting

Question	Answer	Marks	Guidance
13	vitamin B12 / cobalamin – metabolism of amino acids, prevention of megaloblastic anaemia, formation of red blood cells, synthesis of DNA, normal nerve function, normal brain function; vitamin E / tocopherol – antioxidant, destroys free radicals, formation of new blood vessels around damaged areas, functioning of sex organs / reproduction / fertility, healthy skin, maintenance of cell membranes / cellular respiration, protection against heart disease, reduces risk of developing certain cancers; vitamin K / phylloquinone / menaquinone – aids the absorption of calcium in bone structure, coagulation/clotting of blood e.g. helps after injury, important in maintaining vitality and longevity, normal liver functioning, protects against osteoporosis, all newborns given vitamin K injection to prevent VKDB; iron – production of haemoglobin, formation of red blood cells, to transport oxygen around the body, helps convert blood sugar to energy, prevent anaemia; iodide – production of thyroid hormones, functioning of thyroid gland, prevent goitre, energy metabolism, normal skin, nervous system function; phosphorous – works with calcium as a component of bones and teeth, helps function of cell membranes, helps to metabolise fats and proteins to produce energy; low in carbohydrates / sugar / starch – good for low energy dense / weight management diets, helps control blood sugar levels, prevent type 2 diabetes, ideal for coeliacs; *versatility [max 8]* can be incorporated into lots of dishes both sweet and savoury or be cooked in different ways to use as a main dish, breakfast, dessert, snack e.g. omelette, scrambled, boiled, poached, pickled etc.; can create colour / improve appearance by using as a glaze on scones, pastry, bread products;		Shows good knowledge and understanding of versatility of eggs Should name function, give an explanation and at least one example for full mark If only stated a way in which eggs are versatile with no example then ● and allow 1 mark for 2 functions

Question	Answer	Marks	Guidance
13	can be used to coat food for frying forming a protective layer on the outside which sets and holds the food together e.g. fish fingers, Scotch eggs, chicken portions; can be used to bind two or more ingredients together e.g. fish cakes, stuffing, beef burger, croquettes, rissoles; can be used as an emulsifier which enables oil and water to be mixed together without separating e.g. salad dressing, rich cakes, Hollandaise sauce, Bearnaise sauce, Aioli, carbonara, ice cream; can be used as a raising agent to aerate a dish as air can be whipped into egg white or whole egg to form a foam e.g. meringue, whisked sponge, Swiss roll, choux pastry, souffle, mousse; can be used as a garnish to improve appearance by slicing or chopping e.g. salad, dressed crab, ramen, nicoise salad, nasi goreng; can be used instead of meat or fish in a main meal to provide for vegetarians / lacto-ovo vegetarian e.g. savoury flan, tortilla, frittata, omelette, curried egg; can be used to form structure / set or thicken a mixture due to coagulation properties e.g. Victoria sponge, egg custard, lemon curd, quiche; can be used to enrich a dish by providing flavour / colour / texture / extra nutrients e.g. egg fried rice, mashed potato, milk pudding, sauce, brioche; easy to eat and digest so good for elderly, convalescents, young children who may have a weaker digestive system; fresh eggs have a longer shelf life than fresh meat / fresh fish / fresh milk / fresh cheese and do not need refrigeration / special storage conditions; eggs are cheaper in comparison to meat / fish so more affordable for those on limited budget;		correct terminology used where appropriate comments precise and relevant information should be presented in a clear and organised way